PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 2 · Version A

21 questions · 90 minutes · original practice exam, not an official exam

Name _____ NetID _____ Section _____

Closed book. Calculators allowed. Use the official PHY 2111 formula sheet. Take $g = 9.81\ m/s^2$.
MC5 questions have five choices (6 points), MC3 three choices (3 points), MC2 two choices (2 points).
Mark one answer per question. Answers and worked solutions are in the separate solutions file.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 5 kg crate starts from rest and is pushed 8 m across a level floor by a 40 N force directed 20° below the horizontal. The coefficient of kinetic friction is 0.3.

1) How much work does the pushing force do on the crate? **MC5**

- **a.** 281 J
- **b.** 109 J
- **c.** 301 J
- **d.** 320 J
- **e.** 151 J

2) What is the speed of the crate after it has moved 8 m? **MC5**

- **a.** 5.48 m/s
- **b.** 11 m/s
- **c.** 8.56 m/s
- **d.** 7.75 m/s
- **e.** 5.5 m/s

3) How much work does the normal force do on the crate? **MC3**

- **a.** A negative amount
- **b.** A positive amount
- **c.** Zero

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A toy launcher has a spring with k = 800 N/m. It is compressed 0.15 m and fires a 0.2 kg ball straight up. Treat the launch point as the release point and ignore air resistance.

4) What is the launch speed of the ball? MC5

a. 9.49 m/s
b. 4.74 m/s
c. 6.71 m/s
d. 30 m/s
e. 13.4 m/s

5) How high above the launch point does the ball rise? MC5

a. 6.12 m
b. 9.17 m
c. 3.06 m
d. 4.59 m
e. 2.29 m

6) If the spring were compressed twice as far, the maximum height would be multiplied by MC3

a. √2
b. 4
c. 2

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 2 kg ball on a 1.5 m string is pulled aside until the string makes 60° with the vertical, then released from rest.

7) What is the speed of the ball at the lowest point? MC5

a. 1.92 m/s
b. 3.84 m/s
c. 5.42 m/s
d. 4.6 m/s
e. 2.71 m/s

8) What is the tension in the string at the lowest point? MC5

a. 39.2 N
b. 58.9 N
c. 19.6 N
d. 9.81 N
e. 29.4 N

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1200 kg car accelerates from rest to 25 m/s in 9 s on a level road. Ignore friction and air resistance.

9) What average power does the engine deliver? **MC5**

- **a.** 375 kW
- **b.** 3.33 kW
- **c.** 30 kW
- **d.** 41.7 kW
- **e.** 83.3 kW

10) To reach 50 m/s from rest in the same time, the average power would need to be multiplied by **MC3**

- **a.** 4
- **b.** 2
- **c.** 8

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1500 kg car moving east at 20 m/s rear-ends a 2500 kg truck at rest. The two vehicles lock together.

11) What is their speed just after the collision? **MC5**

- **a.** 20 m/s
- **b.** 12 m/s
- **c.** 7.5 m/s
- **d.** 5 m/s
- **e.** 10 m/s

12) How much kinetic energy is lost in the collision? **MC5**

- **a.** 75 kJ
- **b.** 300 kJ
- **c.** 0 kJ
- **d.** 113 kJ
- **e.** 188 kJ

13) During the collision, the force of the truck on the car is **MC3**

- **a.** larger than the force of the car on the truck.
- **b.** equal in size to the force of the car on the truck.
- **c.** smaller than the force of the car on the truck.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.5 kg ball moving at 4 m/s makes a head-on elastic collision with a 1.5 kg ball at rest.

14) What is the velocity of the 0.5 kg ball after the collision? **MC5**

- **a.** 0
- **b.** 2.0 m/s forward
- **c.** 4.0 m/s backward
- **d.** 1.0 m/s forward
- **e.** 2.0 m/s backward

15) What is the speed of the 1.5 kg ball after the collision? **MC5**

- **a.** 2 m/s
- **b.** 1 m/s
- **c.** 1.33 m/s
- **d.** 2.67 m/s
- **e.** 4 m/s

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.15 kg baseball arrives at a bat at 35 m/s and leaves in the opposite direction at 45 m/s. The ball is in contact with the bat for 1.2 ms.

16) What is the magnitude of the average force of the bat on the ball? **MC5**

- **a.** 4380 N
- **b.** 5630 N
- **c.** 1.00×10^4 N
- **d.** 12 N
- **e.** 1250 N

17) The impulse on the ball points **MC2**

- **a.** in the direction of the ball's final velocity.
- **b.** in the direction of the ball's initial velocity.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A ceiling fan starts from rest and speeds up uniformly to 900 rpm in 6.0 s. Its blades are 0.40 m long.

18) What is the angular acceleration of the fan? **MC5**

- **a.** 7.85 rad/s²
- **b.** 94.2 rad/s²
- **c.** 15.7 rad/s²
- **d.** 2.5 rad/s²
- **e.** 150 rad/s²

19) How many revolutions does the fan make during these 6.0 s? MC5

a. 22.5 rev
b. 7.2 rev
c. 45 rev
d. 90 rev
e. 283 rev

20) Compared with the blade tip, a point halfway along the blade has MC3

a. the same angular speed and half the linear speed.
b. half the angular speed and the same linear speed.
c. the same angular speed and the same linear speed.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A 2.0 kg uniform rod 1.2 m long has a 0.50 kg point mass attached at each end.

21) What is the moment of inertia about an axis through the center of the rod, perpendicular to it? MC5

a. 0.96 kg·m²
b. 0.6 kg·m²
c. 1.2 kg·m²
d. 0.24 kg·m²
e. 0.36 kg·m²
